

Aufgabe 1: (20 Punkte)

1.

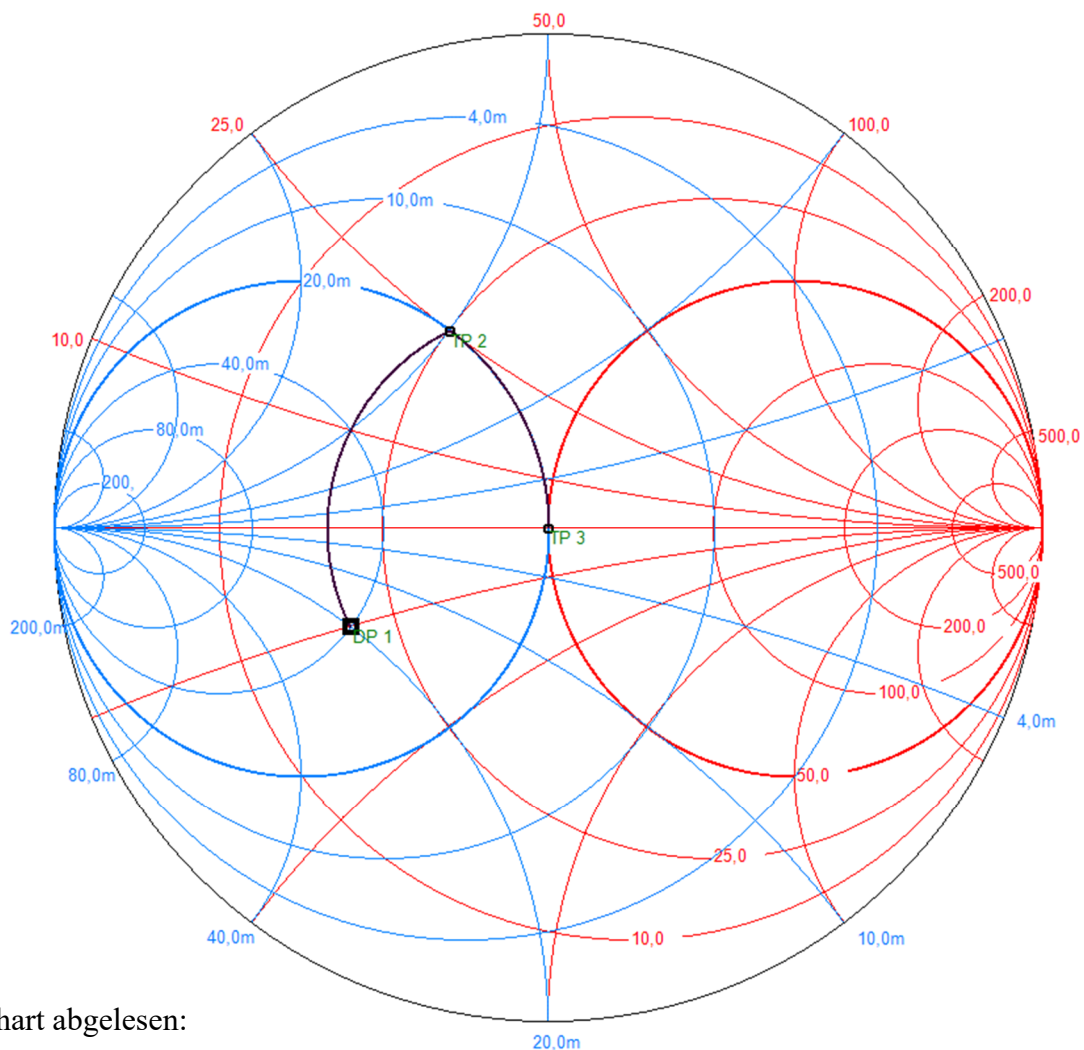
$$\underline{Z}_1 = \frac{R_1 \cdot \frac{1}{j\omega C_1}}{R_1 + \frac{1}{j\omega C_1}} = \frac{25\Omega \cdot \frac{1}{j2\pi \cdot 2,0 \cdot 10^9 \frac{1}{s} \cdot 1,6 \cdot 10^{-12} \frac{As}{V}}}{25\Omega + \frac{1}{j2\pi \cdot 2,0 \cdot 10^9 \frac{1}{s} \cdot 1,6 \cdot 10^{-12} \frac{As}{V}}} = (20 - j10)\Omega$$

$$\text{normiert: } \frac{\underline{Z}_1}{Z_L} = \frac{(20 - j10)\Omega}{50\Omega} = 0,4 - j0,2 \rightarrow \text{ins Smith-Chart eintragen}$$

- oder -

$$\underline{Y}_1 \cdot Z_L = \frac{Z_L}{R_1} + jZ_L \cdot \omega C_1 = 2 + j \rightarrow \text{ins Leitwertdiagramm}$$

2P



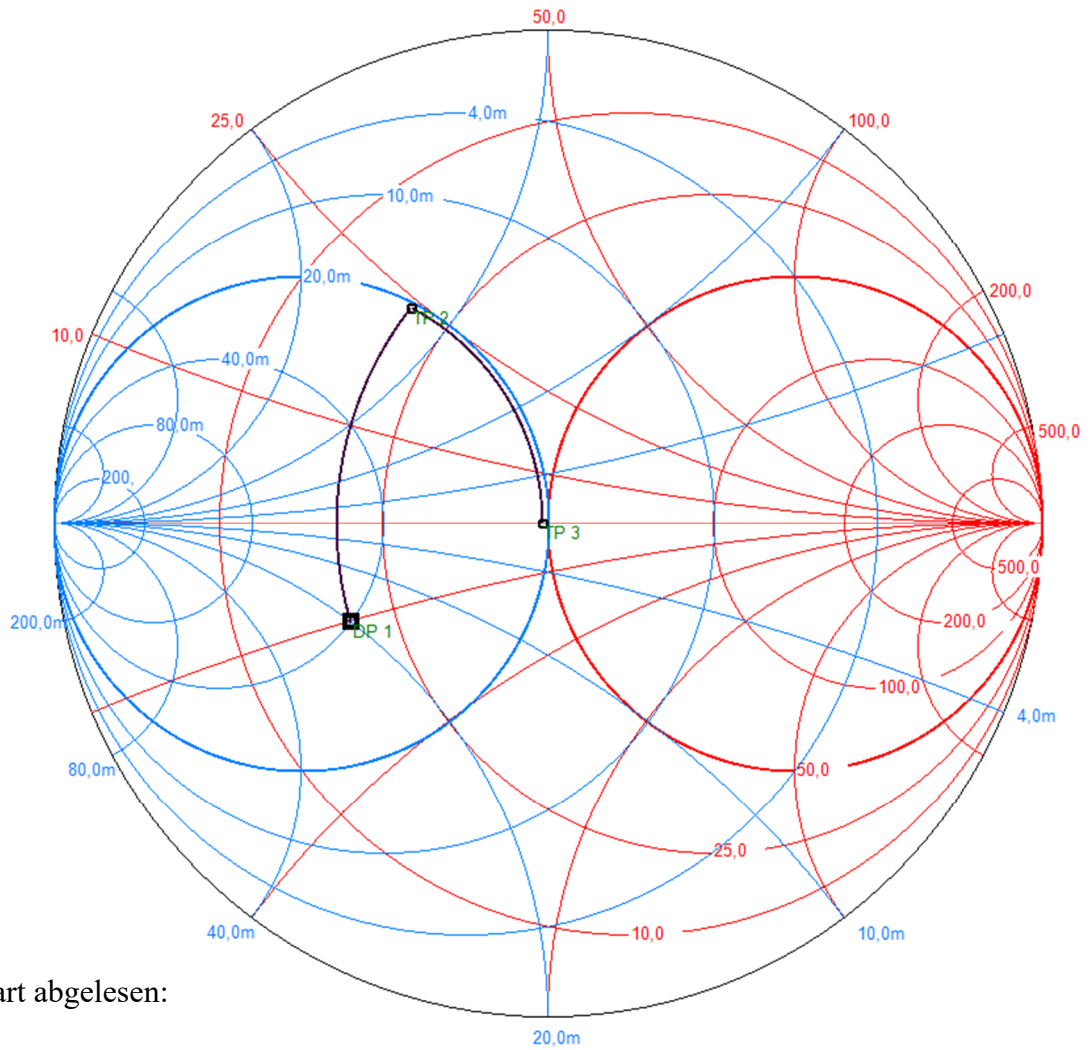
aus Smith-Chart abgelesen:

$$\frac{l}{\lambda} = 0,088 - 0,463 + 0,5 = 0,125 \quad \lambda = \frac{c_0}{\sqrt{\epsilon_r} \cdot f} = \frac{3 \cdot 10^8 \frac{m}{s}}{\sqrt{4} \cdot 2,0 \cdot 10^9 \frac{1}{s}} = 75\text{mm}$$

$$\underline{l} = 0,125 \cdot 75\text{mm} = \underline{\underline{9,4\text{mm}}}$$

$$\frac{X_C}{Z_L} = 1 \rightarrow \omega C = \frac{1}{Z_L} \quad C = \frac{1}{\omega Z_L} = \frac{1}{2\pi \cdot 2,0 \cdot 10^9 \frac{1}{s} \cdot 50 \frac{V}{A}} = \underline{\underline{1,6\text{pF}}}$$

2. Induktivität



aus Smith-Chart abgelesen:

$$\frac{X_L}{Z_L} = 0,49 - (-0,2) = 0,69 \quad \rightarrow \quad \underline{\underline{L}} = \frac{0,69 \cdot Z_L}{\omega} = \frac{0,69 \cdot 50 \frac{\text{V}}{\text{A}}}{2\pi \cdot 2,0 \cdot 10^9 \frac{1}{\text{s}}} = \underline{\underline{2,75 \text{nH}}}$$

dann aber:

$$B_C \cdot Z_L = 1,21 \quad \rightarrow \quad \underline{\underline{C}} = \frac{1,21}{\omega \cdot Z_L} = \frac{1,21}{2\pi \cdot 2,0 \cdot 10^9 \frac{1}{\text{s}} \cdot 50 \frac{\text{V}}{\text{A}}} = \underline{\underline{1,9 \text{pF}}}$$

Aufgabe 2:

(12 Punkte)

1. $C = 5\text{dB} \rightarrow k = \sqrt{10^{\frac{5\text{dB}}{10\text{dB}}}} = 0,5623$

$$Z_{L1} = \frac{Z_L}{k} = \frac{50\Omega}{0,5623} = 88,9\Omega$$
$$Z_{L2} = \frac{Z_L}{\sqrt{1-k^2}} = \frac{50\Omega}{\sqrt{1-0,5623^2}} = 60,5\Omega$$

aus Diagramm: $Z_{L1} \approx 90\Omega \rightarrow w_1/h (\epsilon_r = 6) = 0,4 \rightarrow \underline{\underline{w_1 = 0,4\text{mm}}}$

$$Z_{L2} \approx 60\Omega \rightarrow w_2/h (\epsilon_r = 6) = 1,1 \rightarrow \underline{\underline{w_2 = 1,1\text{mm}}}$$

2. aus Diagramm: $\epsilon_{r,\text{eff}} \approx 4$

$$\lambda = \frac{c_0}{\sqrt{\epsilon_{r,\text{eff}}} \cdot f} = \frac{3 \cdot 10^8 \frac{\text{m}}{\text{s}}}{\sqrt{4} \cdot 3,0 \cdot 10^9 \frac{1}{\text{s}}} = 50\text{mm}$$

Umfang Rat-Race-Koppler:

$$U \approx 2 \cdot \frac{3\lambda}{4} = \frac{6\lambda}{4} = 1,5\lambda = 75\text{mm}$$

$$\underline{\underline{D}} = \frac{U}{\pi} = \frac{75\text{mm}}{\pi} = \underline{\underline{24\text{mm}}}$$

Aufgabe 3:

(14 Punkte)

$$1. \quad \underline{s}_{11} = \frac{\underline{Z}_E - \underline{Z}_L}{\underline{Z}_E + \underline{Z}_L} = \frac{30\Omega + j20\Omega - 50\Omega}{30\Omega + j20\Omega + 50\Omega} = 0,3430 \cdot e^{j121^\circ} = -0,177 + j0,294$$

$$\underline{s}_{22} = \frac{\underline{Z}_A - \underline{Z}_L}{\underline{Z}_A + \underline{Z}_L} = \frac{10\Omega - j30\Omega - 50\Omega}{10\Omega - j30\Omega + 50\Omega} = 0,7454 \cdot e^{-j117^\circ} = -0,338 - j0,664$$

$$\underline{s}_{21} = 20 \cdot e^{j90^\circ} \quad \underline{s}_{12} = 0$$

2.

$$\underline{\underline{v_{Tr,max}}} = 10\text{dB} \cdot \lg \left(|\underline{s}_{21}|^2 \cdot \frac{1}{1 - |\underline{s}_{11}|^2} \cdot \frac{1}{1 - |\underline{s}_{22}|^2} \right) = 10\text{dB} \cdot \lg \left(20^2 \cdot \frac{1}{1 - 0,3^2} \cdot \frac{1}{1 - 0,7^2} \right) = \underline{\underline{29,4\text{dB}}}$$

3.

$$K = \frac{1 - |\underline{s}_{11}|^2 - |\underline{s}_{22}|^2 + |\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}|^2}{2|\underline{s}_{12}\underline{s}_{21}|} = \frac{1 - 0,3^2 - 0,7^2 + |0,3 \cdot (-0,7) - 0,01 \cdot 20|^2}{2 \cdot 0,01 \cdot 20} =$$

$$= 1,47 > 1$$

$$B = 1 + |\underline{s}_{11}|^2 - |\underline{s}_{22}|^2 - |\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}|^2 = 1 + 0,3^2 - 0,7^2 - |0,3 \cdot (-0,7) - 0,01 \cdot 20|^2 =$$

$$= 0,432 > 0$$

Transistor ist also absolut stabil!

Aufgabe 4:

(20 Punkte)

1. Rauschzahlen als entlogarithmierte Werte:

$$F_1 = 10^{\frac{2\text{dB}}{10\text{dB}}} = 1,58 \quad F_2 = 4,0 \quad F_3 = 2,0 \quad F_L = \frac{1}{G_L} = 10$$

$$\begin{aligned} \underline{\underline{F_{ges}}} &= 1 + (F_1 - 1) + \frac{F_L - 1}{G_1} + \frac{F_2 - 1}{G_1 \cdot G_L} + \frac{F_3 - 1}{G_1 \cdot G_L \cdot G_2} = \\ &= 1 + (1,58 - 1) + \frac{10 - 1}{10} + \frac{4 - 1}{10 \cdot \frac{1}{10}} + \frac{2 - 1}{10 \cdot \frac{1}{10} \cdot 100} = 5,49 \hat{=} \underline{\underline{7,4\text{dB}}} \end{aligned}$$

$$2. \quad \frac{P_{R,aus}}{P_{R,ein}} = F_{ges} \cdot G_{ges} \quad \frac{P_{S,aus}}{P_{S,ein}} = G_{ges}$$

$$P_{R,ein} = k \cdot T_A \cdot B = 1,38 \cdot 10^{-23} \frac{\text{Ws}}{\text{K}} \cdot 40\text{K} \cdot 20 \cdot 10^6 \frac{1}{\text{s}} = 1,1 \cdot 10^{-14} \text{W}$$

$$P_{R,aus} = F_{ges} \cdot G_{ges} \cdot P_{R,ein} = 5,49 \cdot 10^{\frac{10\text{dB} - 10\text{dB} + 20\text{dB} + 6\text{dB}}{10\text{dB}}} \cdot 1,1 \cdot 10^{-14} \text{W} = 2,40 \cdot 10^{-11} \text{W}$$

$$P_{S,aus,min} = 10 \cdot P_{R,aus} = 2,40 \cdot 10^{-10} \text{W}$$

$$\underline{\underline{P_{S,ein,min}}} = \frac{P_{S,aus,min}}{G_{ges}} = \frac{2,40 \cdot 10^{-10} \text{W}}{10^{2,6}} = \underline{\underline{6,0 \cdot 10^{-13} \text{W}}}$$

oder:

$$\left(\frac{S}{N} \right)_{ein} = F_{ges} \cdot \left(\frac{S}{N} \right)_{aus} \hat{=} 7,4\text{dB} + 10\text{dB} = 17,4\text{dB}$$

$$\underline{\underline{P_{S,ein,min}}} = 10^{\frac{16,7\text{dB}}{10\text{dB}}} \cdot P_{R,ein} = 10^{1,74} \cdot 1,1 \cdot 10^{-14} \text{W} = \underline{\underline{6,0 \cdot 10^{-13} \text{W}}}$$

3. Anordnung in Reihenfolge zunehmender Rauschzahl, Leitung zuletzt:

Antenne – V1 – V3 – V2 – Leitung

$$\begin{aligned} \underline{\underline{F_{ges}}} &= 1 + (F_1 - 1) + \frac{F_3 - 1}{G_1} + \frac{F_2 - 1}{G_1 \cdot G_3} + \frac{F_L - 1}{G_1 \cdot G_3 \cdot G_2} = \\ &= 1 + (1,58 - 1) + \frac{2 - 1}{10} + \frac{4 - 1}{10 \cdot 4} + \frac{10 - 1}{10 \cdot 4 \cdot 100} = 1,757 \hat{=} \underline{\underline{2,45\text{dB}}} \end{aligned}$$

Aufgabe 5:

(11 Punkte)

1. Leerlauf $\rightarrow \underline{r}_L = 1$

$$\underline{r}_1 = \underline{s}_{11} + \frac{\underline{s}_{12} \cdot \underline{s}_{21} \cdot \underline{r}_L}{1 - \underline{s}_{22} \cdot \underline{r}_L} = j0,5 + \frac{0,8 \cdot 0,8 \cdot 1}{1 - j0,5 \cdot 1} = 0,9131 \cdot e^{j56^\circ} = 0,511 + j0,757$$

$$|\underline{r}_1|^2 = 0,8337$$

$$\underline{\underline{P_Q}} = P_{max} \cdot (1 - |\underline{r}_1|^2) = 10\text{W} \cdot (1 - 0,8337) = \underline{\underline{1,66\text{W}}}$$

2. $\underline{r}_1 = \underline{s}_{11} + \frac{\underline{s}_{12} \cdot \underline{s}_{21} \cdot \underline{r}_L}{1 - \underline{s}_{22} \cdot \underline{r}_L} = 0$ für Anpassung

$$-\underline{s}_{11} \cdot (1 - \underline{s}_{22} \cdot \underline{r}_L) = \underline{s}_{12} \cdot \underline{s}_{21} \cdot \underline{r}_L$$

$$-\underline{s}_{11} + \underline{s}_{11} \cdot \underline{s}_{22} \cdot \underline{r}_L = \underline{s}_{12} \cdot \underline{s}_{21} \cdot \underline{r}_L$$

$$\underline{r}_L \cdot (\underline{s}_{11} \cdot \underline{s}_{22} - \underline{s}_{12} \cdot \underline{s}_{21}) = \underline{s}_{11}$$

$$\underline{\underline{\underline{r}_L}}} = \frac{\underline{s}_{11}}{\underline{s}_{11} \cdot \underline{s}_{22} - \underline{s}_{12} \cdot \underline{s}_{21}} = \frac{j0,5}{j0,5 \cdot j0,5 - 0,8 \cdot 0,8} = \underline{\underline{0,5618 \cdot e^{-j90^\circ}}}$$

damit: Anpassung, d.h. $P = P_{max} = 10\text{W}$

Aufgabe 6:

(13 Punkte)

1.

$$C = 20\text{dB} \rightarrow c = 0,01 \quad \beta l = \frac{360^\circ}{\lambda} \cdot \frac{\lambda}{20} = 18^\circ$$

$$c = |\underline{s}_{41}|^2 = \frac{k_L^2 \cdot \sin^2 \beta l}{(1 - k_L^2) \cdot \cos^2 \beta l + \sin^2 \beta l} = \frac{k_L^2 \cdot \sin^2 \beta l}{1 - k_L^2 \cdot \cos^2 \beta l}$$

$$k_L^2 \cdot \sin^2 \beta l = c - c \cdot k_L^2 \cdot \cos^2 \beta l$$

$$k_L^2 \cdot (\sin^2 \beta l + c \cdot \cos^2 \beta l) = c$$

$$\underline{\underline{k_L}} = \sqrt{\frac{c}{\sin^2 \beta l + c \cdot \cos^2 \beta l}} = \sqrt{\frac{0,01}{\sin^2 18^\circ + 0,01 \cdot \cos^2 18^\circ}} = \underline{\underline{0,31}}$$

2. vierfache Frequenz: Länge des Kopplers = $4 \cdot \lambda/20 = \lambda/5$

$$\beta l = \frac{360^\circ}{\lambda} \cdot \frac{\lambda}{5} = 72^\circ$$

$$\underline{\underline{c}} = |\underline{s}_{41}|^2 = \frac{k_L^2 \cdot \sin^2 \beta l}{1 + k_L^2 \cdot \cos^2 \beta l} = \frac{0,31^2 \cdot \sin^2 72^\circ}{1 + 0,31^2 \cdot \cos^2 72^\circ} = 8,77 \cdot 10^{-2} \triangleq \underline{\underline{10,6\text{dB}}}$$